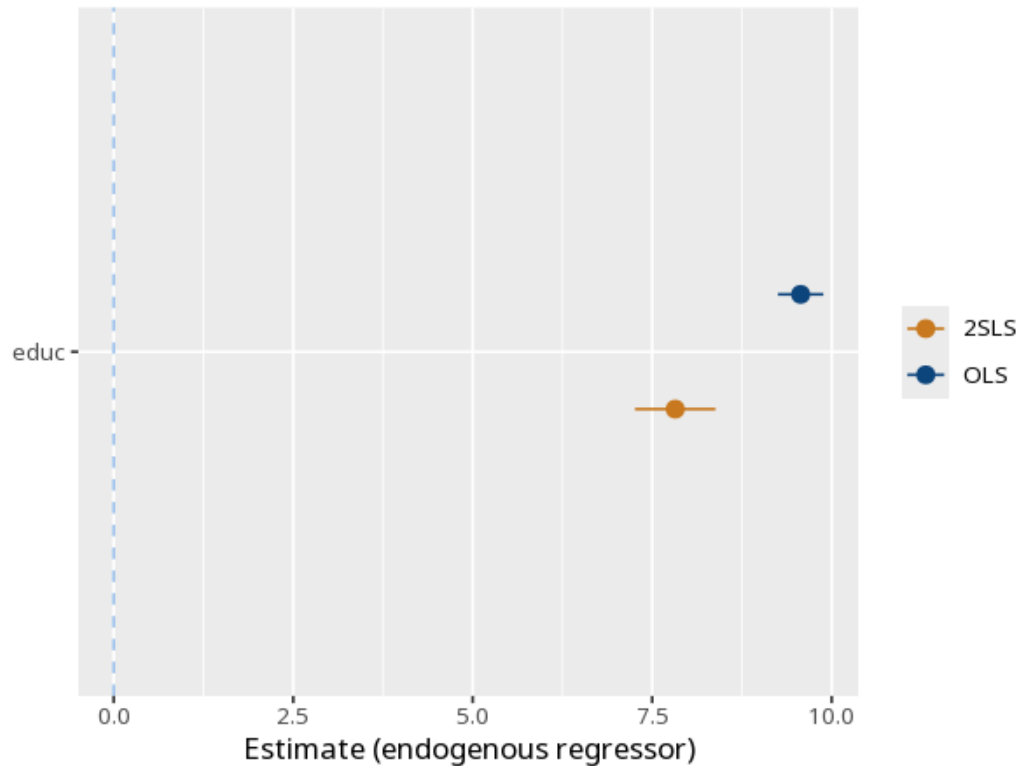


1 Instrumental variables (IV / 2SLS)

Endogenous	Instrument	Coef.	SE	Partial F	p
educ	educ_iv	1.16	0.06	438.50	<.001
OLS		2SLS			
(Intercept)	178.90 (2.41) [174.16, 183.65]	202.74 (4.10) [194.65, 210.83]			
educ	9.57 (0.16) [9.25, 9.88]	7.82 (0.28) [7.26, 8.38]			
exper	-0.02 (0.05) [-0.11, 0.08]	-0.02 (0.06) [-0.14, 0.09]			
Num.Obs.		200			
RMSE		6.67			
F		373.44			
Weak-instrument F = 438.50, p < .001 (rule of thumb: F > 10)					
Wu-Hausman = 1022.75, p < .001					
Sargan: — (just-identified)					



Check the first-stage F first (rule of thumb > 10 = strong instruments). Then read the 2SLS B for the endogenous predictor as the causal effect, with p/CI. This causal reading rests on the instrument being valid — relevant and affecting the outcome only through the endogenous predictor (the exclusion restriction). That is an assumption you must justify on theory; the Sargan test only checks over-identification (extra instruments), not validity, and is unavailable for just-identified models. Method: R `ivreg` / AER package; weak-instrument F (Stock & Yogo, 2005). Your significance threshold (α) is 0.05.

The 2SLS estimate for `educ` was $B=7.82$, 95% CI $[7.26, 8.38]$, $p < .001$ (first-stage $F=438.50$).

Statistical basis: Two-stage least squares (2SLS) estimation (Fox J, Kleiberg C, Zeileis A (2024). `ivreg`: Instrumental-Variables Regression by 2SLS, 2SM, or 2SMM, with Diagnostics. R package.); Weak-instrument first-stage F rule of thumb (Stock, J. H., Yogo, M. (2005). "Testing for Weak Instruments in Linear IV Regression." In *Identification and Inference for Econometric Models* (Andrews & Stock, eds.), Cambridge University Press, 80-108.); Heteroscedasticity-robust standard errors (Zeileis A, Köll S, Graham N (2020). "Various Versatile Variances: An Object-Oriented Implementation of Clustered Covariances in R." *Journal of Statistical Software*, 95(1), 1-36.); Wu-Hausman endogeneity test (Wu, D. M. (1973). "Alternative tests of independence between stochastic regressors and disturbances." *Econometrica*, 41(4), 733-750.); Sargan over-

identification test (Sargan, J. D. (1958). "The estimation of economic relationships using instrumental variables." *Econometrica*, 26(3), 393-415.).